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## NUTRAVIT SYRUP

VINUTxx-0

**Desription**Brown, translucent, viscous liquid with orange flavor.

### Composition

| | | |
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| Per 5ml: | Vitamin A | 4000 USP Units |
| | Vitamin D | 400 USP Units |
| | Ascorbic Acid | 17.5 mg |
| | Thiamine HCl | 2.0 mg |
| | Riboflavine | 2.0 mg |
| | Pyridoxine HCl | 2.0 mg |
| | Calcium Pantothenate | 3.25 mg |
| | Cyanocobalamin | 1.5 mcg |
| | Nicotinamide | 10.0 mg |
| | L-Lysine HCl | 100.0 mg |

### Pharmacodynamics

**Vitamin A**
Vitamin A is essential for normal function of the retina; in the form of retinal, it combines with opsin (red pigment in the retina) to form rhodpsin (visual purple) which is necessary for visual adaptation to darkness.

Other forms (retinol, retinoic acid) are necessary for growth of bone reproduction, and embryonic development, and to preserve epithelial integrity; may act as a cofactor in biochemical reactions.

**Vitamin D**
Vitamin D is essential for promoting absorption and utilization of calcium and phosphate. Along with parathyroid hormone and calcitonin, it regulates serum calcium levels by increasing serum calcium and phosphate levels as needed.

Cholecalciferol is converted to calcifediol in the liver and to calcitriol in the kidney.

The metabolic activity of calcifediol appears to be not only to its conversion to other metabolites but also to its intrinsic vitamin D activity. Serum calcium levels are increased, alkaline phosphatase and parathyroid hormone levels are decreased, subperiosteal bone resorption is decreased and histological signs of hyperparathyroid bone disease and mineralization defects are decreased.

Calcitriol, along with parathyroid hormone, regulates the transfer of calcium ion from bone to the extracellular fluid thereby effecting calcium homeostasis in the extracellular fluid.

**Cyanocobalamin**
Cyanocobalamin is a synthetic form of Vitamin B12. It acts as a coenzyme for various metabolic functions, including fat and carbohydrate metabolism and protein synthesis. It is essential for growth, cell replication, hematopoiesis, and nucleoprotein and myelin synthesis, largely due to its effects on metabolism of methionine, folic acid, and malonic acid.

**Pyridoxine**
In the body, pyridoxine is converted to pyridoxal phosphate, which is a coenzyme for a wide variety of metabolic transformations, including decarboxylation, transamination, racemization, and the metabolism of tryptophan, sulphur containing amino-acids, and hydroxyamino acids.

It plays a part in protein metabolism, the synthesis of fat from protein, haemopoiesis, and the nutrition of the skin.

**Thiamine**
Thiamine pyrophosphate, the physiologically active form of thiamine, functions in carbohydrate metabolism as acoenzyme in the decarboxylation of x-ketoacids particularly of pyruvate and x-ketoglutarate.

In the presence of thiamine deficiency pyruvic and lactic acids accumulate in the tissues.

It also acts as a coenzyme in the direct oxidative pathway of glucose metabolism.

**Calcium Pantothenate**
Calcium pantothenate is a synthetic derivative of pantothenic acid. Pantothenic acid, a member of the vitamin B group, is a precursor of coenzyme A.

Coenzyme A is involved in many vital enzymatic reactions transferring a two-carbon compound (the acetyl group) in intermediary metabolism, and is required for various metabolic functions, including the release of energy from carbohydrates, in the degradation and metabolism of fatty acids, and in the synthesis of such compounds as sterols and steroid hormones, porphyrins, and acetylcholine.

It may also be necessary for normal epithelial function.

**Riboflavine**
Riboflavine, a member of the vitamin B group, is inactive until converted to the physiologically active forms, flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD), the coenzyme necessary for tissue respiration.

BACK

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Riboflavine is also required for activation of pyridoxine, and may be involved in maintaining erythrocyte integrity.

**Nicotinamide**
Nicotinamide is converted in the body to nicotinamide adenine dinucleotide (NAD) and nicotinamide adenine dinucleotide phosphate (NADP) which are involved in electron transfer reactions in the respiratory chain. They are also necessary for lipid metabolism and glycogenolysis.

Nicotinamide does not cause vasodilation like nicotinic acid.

**Ascorbic Acid**
Ascorbic acid (or Vitamin C) is essential for the synthesis of collagen and intercellular materials, and hence for the healing of wounds. It is also necessary for the conversion of folic acid to folinic acid. It facilitates the absorption of iron from the gut and influences the formation of haemoglobin and erythrocyte maturation. It is also involved in metabolism of tyrosine, carbohydrates and synthesis of lipids and proteins.

**Lysine**
Lysine is an amino acid which is essential in human nutrition, not synthesized by the human body.

The hydrochloride has been used in the treatment of hypochloraemic alkalosis. Lysine is also used as a dietary supplement. Supplementation of wheat-based foods with lysine improves their protein quality and results in improved growth and tissue synthesis.

### Pharmacokinetics
It is well absorbed orally and is excreted mainly in urine and faeces.

### Indications
As a health supplement

### Contraindications
Not recommended for patients with hypervitaminosis A or D, hypercalcaemia, renal osteodystrophy with hyperphosphatemia and Leber's disease.

### Precautions
Caution in patients with arteriosclerosis, renal, hepatic or cardiac function impairment, hyperphosphatemia, hypersensitivity to effects of vitamin D, gout, haemochromatosis, sideroblastic anaemia, thalassaemia, hyperoxaluria, peptic ulcer, haemophilia, diabetes mellitus, glaucoma and hyperchloraemic acidosis.

### Pregnancy and Lactation
None known

### Main Side/Adverse Effects
Diarrhoea, itching, headache, flushing or redness of skin, nausea or vomiting, gastrointestinal discomfort and yellow discolouration of urine may occur occasionally.

### Drug Interactions
None known

### Overdosage
**Clinical features:**
Headache, nausea, vomiting, abdominal pain, anorexia, diarrhoea, drowsiness, flushing, hypertension, acute renal failure, papilloedema, generalised peeling of the skin, pruritus, polyuria, polydipsia, fatigue, muscle weakness.

**Treatment for overdosage:**
Emesis or gastric lavage if appropriate. Symptomatic and supportive measures.

### Dosage and Administration
Adults: Oral, before or after food, 5ml once daily;or as directed.
Children (4 to 12 years old): Oral, before or after food, 2.5ml to 5ml once daily; or as directed.

The information given here is limited. For further information, consult your doctor or pharmacist.

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| Storage | : | Store below 25°C. Protect from light and freezing. |
| Presentation/Packing | : | Syrup x 60 ml, 100 ml, 120 ml, 125 ml |
| Product Registration Holder | : | **HOVID Pharmacy Sdn. Bhd.,** 121, Jalan Tunku Abdul Rahman (Jalan Kuala Kangsar), 30010 Ipoh, Perak, Malaysia. |
| Manufactured by | : | **HOVID Bhd.,** 121, Jalan Tunku Abdul Rahman (Jalan Kuala Kangsar), 30010 Ipoh, Perak, Malaysia. |

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